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Dynamic Trade and Exchange Rate Adjustments: A Differential-Equation Framework for Open-Economy Macrodynamics

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ABSTRACT

This paper examines the dynamic interplay among the trade balance, exchange-rate adjustment, capital accumulation, and output in an open economy through a continuous-time system of differential equations. The mathematical model is constructed such that the exchange rate responds to trade imbalances, capital evolves through savings and depreciation, and output changes in accordance with productive capacity and the external trade position. The objective of the study is not immediate empirical estimation, but rather a theoretical understanding of the equilibrium structure and local stability within a stylized open economy. Steady-state conditions are derived, a Jacobian matrix is employed to analyze local behavior, and a reduced two-dimensional subsystem is examined to elucidate the relationship between capital and output. The results indicate that, under the hypothesized adjustment rule, the exchange-rate equilibrium is locally stable, whereas broader macroeconomic dynamics depend on the boundary relationship between stabilizing forces and strong feedback effects. The study demonstrates that long-run external adjustment relies not only on exchange-rate movements but also on the strength of domestic accumulation and output response. The framework provides a compact theoretical foundation for analyzing open-economy adjustment and can be extended in future work to incorporate policy interventions, stochastic disturbances, and empirical calibration.

Keywords: Exchange rate, trade balance, capital accumulation, open economy, differential equations.

JEL Classification: F31, F43, C61

INTRODUCTION In an open economy, fluctuations in exchange rates, trade imbalances, Capital formation, and output growth are closely interconnected. Changes in the exchange rate affect export competitiveness and import demand; this, in turn, influences the trade balance, domestic production, and the pace of capital accumulation. For this reason, the adjustment process of an open economy cannot

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be fully understood by examining trade, growth, and macroeconomic dynamics in isolation. A more fruitful approach is to study these variables within an integrated dynamic framework. The theoretical literature has examined international trade, exchange rate adjustment, and economic growth separately. Classical and modern trade theories explain how trade patterns respond to comparative advantage, market structure, and productivity differentials (**Smith, 1776; Ricardo, 1817; Samuelson, 1964; Dixit and Norman, 1980**). Growth theory explains how savings, investment, and depreciation determine the long-run evolution of output and capital (**Solow, 1956; Romer, 1986; Barro and Sala-i-Martin, 1995**). Open-economy macroeconomics links external imbalances to exchange-rate movements and broader macroeconomic adjustments (**Obstfeld and Rogoff, 1996; Feenstra, 2004**). However, for many analytical purposes particularly in the case of developing and externally exposed economies, these mechanisms operate simultaneously rather than independently. This creates a need for a compact model in which the external sector and the domestic accumulation process are studied jointly. The present paper develops such a framework by formulating a continuous-time system of differential equations linking three key variables: the exchange rate, the capital stock, and output. In the proposed model, the exchange rate adjusts in response to trade imbalances, capital evolves through saving and depreciation, and output changes in accordance with the economy's productive capacity and trade position. The objective of the paper is not immediate empirical estimation; rather, it aims to elucidate the internal logic of macroeconomic adjustment within a stylized open economy and to identify the conditions under which the adjustment path remains stable. The paper's contribution is twofold. First, it provides a simple nonlinear framework through which the trade balance and capital accumulation can be analyzed jointly. Second, it identifies a boundary condition that distinguishes stable local dynamics from unstable ones. This boundary holds economic significance, as it demonstrates that macroeconomic adjustment depends not only on external feedback but also on the balance between productive expansion and stabilizing forces such as depreciation and output damping. In this sense, the analysis provides a theoretical foundation for understanding why some open economies gradually converge toward equilibrium, while others experience persistent divergence. The remainder of the paper is organized as follows. The next section presents the model structure and basic assumptions. This is followed by an examination of the complete dynamic system, equilibrium analysis, and local stability results. Subsequently, a reduced two-dimensional subsystem is analyzed to more clearly illustrate the geometry of adjustment. The final section summarizes the main findings and highlights their policy implications.

METHODOLOGY

Model structure

An open economy producing a single composite good is considered. The model is intentionally stylized, but its components are drawn from standard ideas in international trade, macroeconomics, and capital accumulation (**Dixit & Norman, 1980; Feenstra, 2004; Obstfeld & Rogoff, 1996; Solow, 1956**). Let Y(t) denote real output, K(t) denote capital stock, E(t) denote the nominal exchange rate, and X(t) and M(t) denote exports and imports, respectively.

Trade balance

Exports and imports are assumed to depend on the exchange rate according to

$X(E) = X_0 E^{\eta}, \quad M(E) = M_0 E^{\theta}$,------(1)

where η and θ are trade elasticities. This reduced-form specification captures the sensitivity of trade flows to exchange-rate movements (**Dornbusch, Fischer, & Samuelson, 1977; Obstfeld & Rogoff, 1996; Feenstra, 2004**). It is assumed that $\eta > \theta$, so that the export response dominates the import response over the relevant range. Economically, this assumption reflects a situation in which exchange-rate adjustment improves the trade position through a stronger response of exports relative to imports.

The trade balance is therefore

$TB(E) = X_0 E^{\eta} - M_0 E^{\theta}$ -------(2)

Its qualitative shape and the location of the steady-state exchange rate are represented schematically in Figure 1.

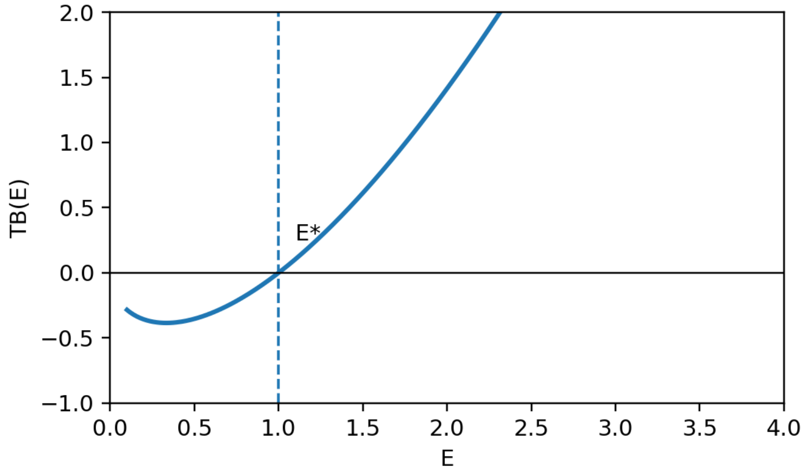

Figure 1. This shows trade-balance function with a unique zero at E*.

Exchange-rate dynamics

The exchange rate is assumed to adjust in response to trade imbalance:

$$\dot{E} = -\alpha\, TB(E) = -\alpha\left(X_0 E^{\eta} - M_0 E^{\theta}\right), \alpha > 0 \quad \text{------------------------------(3)}$$

The negative sign is introduced to represent a stabilizing adjustment mechanism. Under this rule, a trade imbalance generates exchange-rate movement in the direction that tends to reduce that imbalance over time. In other words, the exchange rate behaves here as an adjustment variable rather than as an exogenous policy parameter.

Capital accumulation

Capital evolves according to

$\acute{K} = sY - \delta K$---------(4)

where "s" is the savings rate and δ is the depreciation rate. This equation follows the usual logic of growth theory: output contributes to investment through saving, while depreciation reduces the existing stock of capital (Solow, 1956; Barro & Sala-i-Martin, 1995). The formulation keeps the domestic accumulation mechanism simple so that its interaction with trade adjustment can be seen clearly.

Output dynamics

Output is assumed to change according to

$$\dot{Y} = \gamma\, TB(E) + \beta K - \mu Y ----(5)$$

where γ, β, and μ are positive parameters. The term βK reflects the productive contribution of capital, while γTB(E) captures the influence of the external trade position on domestic output dynamics (**Romer, 1986; Grossman & Helpman, 1991; Obstfeld & Rogoff, 1996**). The term μY represents damping in the output adjustment process. Thus, output is influenced both by internal productive capacity and by the external sector.

RESULTS AND DISCUSSION

Complete system

The full model takes the form

$$\left.\begin{aligned} \dot{E} &= -\alpha(X_0 E^{\eta} - M_0 E^{\theta}) \\ \acute{K} &= sY - \delta K \\ \dot{Y} &= \gamma(X_0 E^{\eta} - M_0 E^{\theta}) + \beta K - \mu Y \end{aligned}\right\} \quad \text{---------------------(6)}$$

This three-dimensional nonlinear system captures the interaction among exchange-rate adjustment, trade balance, capital formation, and output. The structure is simple enough for theoretical analysis, yet rich enough to show how external and internal adjustment mechanisms may reinforce or offset one another.

Equilibrium analysis

An equilibrium point satisfies Ė = Ḱ = Ẏ = 0. Studying these conditions helps identify the long-run structure implied by the model.

Exchange-rate equilibrium: From Ė = 0, one obtains

$$X_0 E^{\eta} = X_0 E^{\theta} ----(7)$$

Hence,

$$E^{*} = \left(\frac{M_0}{X_0}\right)^{\frac{1}{\eta - \theta}} --- (8)$$

Thus, the steady-state exchange rate is determined by the relative scale parameters of exports and imports together with their elasticities. The result shows that long-run exchange-rate balance depends not merely on the size of trade flows, but also on how strongly those flows respond to exchange-rate movement.

As Capital equilibrium: From Ḱ = 0,we get

$$K^{*} = \left(\frac{S}{\delta}\right) Y^{*} --- (9)$$

This indicates that steady-state capital is proportional to steady-state output, with the proportionality determined by the balance between saving and depreciation.

Output equilibrium: Substituting the capital equilibrium condition into the output equation gives

$$0 = \gamma TB(E^{*}) + \beta K^{*} - \mu Y^{*} --- (10)$$

Since TB(E*) = 0, then we get

$$\beta K^{*} = \mu Y^{*} ----(11)$$

Using $\mathrm{K}^{*} = \left(\frac{\mathrm{s}}{\delta}\right) \mathrm{Y}^{*}$, we obtains

$$\beta \left(\frac{S}{\delta}\right) Y^{*} = \mu Y^{*} ----(12)$$

Therefore,

$$Y^{*} = 0 \quad or \quad \left(\frac{\beta_s}{\delta}\right) = \mu ----(13)$$

The second relation identifies a critical threshold in the parameter space for the existence of non-zero steady-state output. Economically, it implies that productive expansion through capital formation must be balanced by output damping if a non-trivial steady state is to exist.

Local stability analysis

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To study local behaviour near equilibrium, the Jacobian matrix is

$$J = \begin{pmatrix} -\alpha TB'(E) & 0 & 0 \\ 0 & -\delta & s \\ \gamma TB'(E) & \beta & -\mu \end{pmatrix} --(14)$$

Where,

$$TB'(E) = X_0\eta E^{\eta-1} - M_0\,\theta E^{\theta-1} --(15)$$

At equilibrium,

$$TB'(E^*) = E^{\theta-1}\left(X_0\eta E^{\eta-\theta} - M_0\theta\right) --(16)$$

Using the equilibrium condition$X_0E^{\eta-\theta} = M_0$, one gets

$$TB'(E^*) = E^{\theta-1}M_0\,(\eta - \theta) --- (17)$$

Since $\eta > \theta$, it follows that$TB'(E^*) > 0$. Therefore, the exchange-rate component contributes to the eigenvalue

$$-\alpha TB'(E^*) < 0 ----(18)$$

Thus, under the assumed adjustment mechanism, the exchange-rate equilibrium is locally stable. This means that small deviations from the steady-state exchange rate tend to be corrected over time rather than amplified. A conceptual convergence path of the exchange rate is shown in Figure-2.

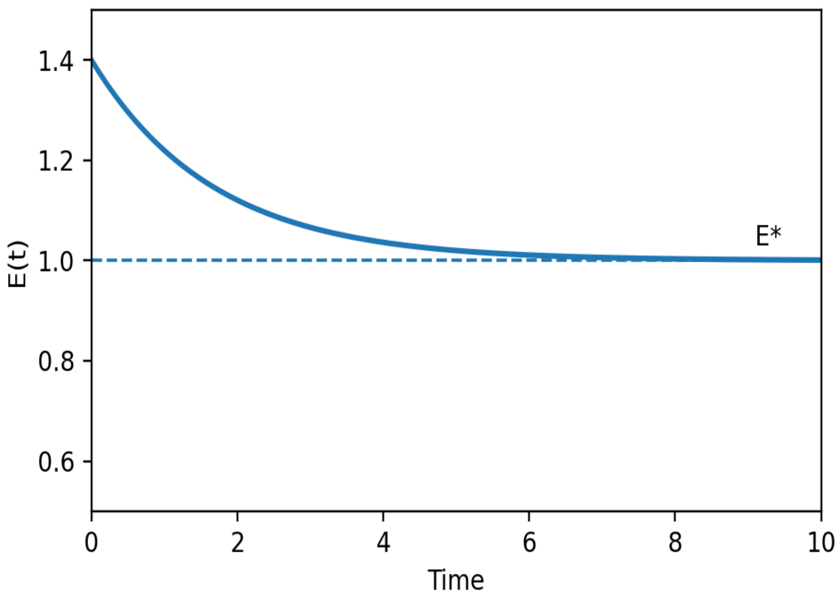

Figure 2.

Conceptual path of the exchange rate converging toward its steady-state value.

The remaining local behaviour depends on the interaction between capital and output, which is examined more clearly through a reduced subsystem.

Reduced two-dimensional subsystem

If $E = E^*$ is fixed, the model reduces to

$\acute{K} = sY - \delta K$
$\dot{Y} = \beta K - \mu Y$ ------------(19)

The nullclines are

$\acute{K} = 0 \Rightarrow K = \left(\frac{s}{\delta}\right) Y$----------(20)

$\dot{Y} = 0 \Rightarrow Y = \left(\frac{\beta}{\mu}\right) K$----------(21)

The geometric interpretation of these nullclines is Shows in Figure 3.

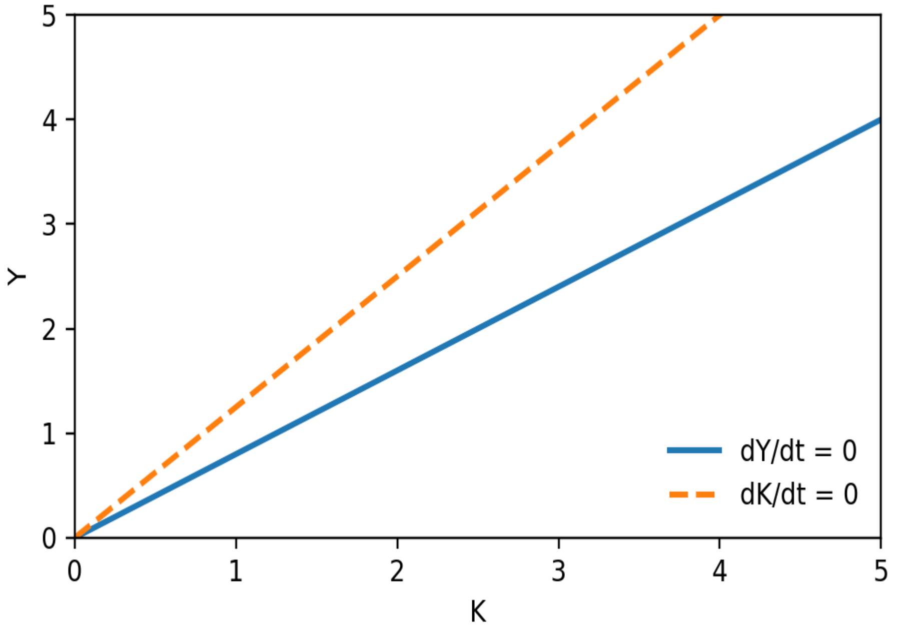

Figure 3. Phase diagram of the (K, Y) subsystem.

The Jacobian matrix of this subsystem is

$$J_{KY} = \begin{pmatrix} -\delta & s \\ \beta & -\mu \end{pmatrix} ----(22)$$

The equilibrium is locally asymptotically stable if

$\delta + \mu > 0, \quad \delta\mu - s\beta > 0$------(23)

Since, $\delta > 0$ and $\mu > 0$, the first inequality already holds. The key requirement is therefore

$\delta\mu > s\beta$--------(24)

This condition shows that convergence depends on whether depreciation and output damping are strong enough to offset the reinforcing effects of savings and capital productivity. In economic terms, the condition separates a regime of controlled adjustment from one in which the feedback between output and capital becomes too strong for local stability.

Dynamic behaviour and threshold effects

The reduced subsystem makes the role of parameters especially transparent. If

$$\delta\mu > s\beta$$

small deviations from equilibrium decay over time. If instead

$$\delta\mu < s\beta ------------ (25)$$

the feedback from savings and capital productivity dominates the stabilising terms, and the system does not return locally to equilibrium.

A critical case arises when

$$\delta\mu = s\beta ------- (26)$$

At this point, the determinant of the Jacobian becomes zero. This marks a threshold at which the qualitative nature of nearby trajectories may change. In that sense, the model reaches a boundary between different local dynamic regimes.

Economically, this result is important because it shows that long-run adjustment depends on the balance between two groups of forces. On one side are depreciation and output damping, both of which work toward stability. On the other side are saving and capital-driven expansion, which may strengthen feedback and move the system away from equilibrium if they become too strong. The threshold therefore has a direct interpretation: it identifies the point beyond which domestic accumulation no longer supports stable macroeconomic adjustment.

CONCLUSIONS AND POLICY IMPLICATIONS

This research paper develops a framework based on differential equations to analyze the joint dynamic behavior of the exchange rate, trade balance, capital accumulation, and output within an open economy. The analysis reveals that all these variables are interconnected through a dynamic adjustment mechanism: external imbalances influence the exchange rate; the exchange rate affects trade outcomes; and domestic output evolves in tandem with capital accumulation. Despite its simple structure, the model yields a meaningful equilibrium configuration and provides clear stability conditions. The results indicate that, under the specified adjustment rules, the exchange rate equilibrium is locally stable. However, the dynamic behavior of the overall economy depends primarily on the interplay between capital accumulation and output adjustment. A reduced subsystem highlights a threshold condition that determines whether local convergence occurs. When stabilizing factors—such as exchange rate depreciation and the damping of output fluctuations—are more potent than the reinforcing effects of savings and capital productivity, the system converges toward equilibrium following minor disturbances. Conversely, when these reinforcing effects become dominant, the adjustment process becomes unstable. The central conclusion of this model is that external adjustment alone is insufficient to ensure macroeconomic stability. Although adjustments in the exchange rate can mitigate trade imbalances, the long-term trajectory of the economy hinges on the sensitivity of domestic output response, the efficiency with which savings are transformed into productive capital, and the effectiveness with which destabilizing feedback mechanisms are controlled. Therefore, achieving stability in an open economy necessitates coordination between external sector adjustments and domestic growth processes. From a policy perspective, this study suggests that economies facing persistent external imbalances should not rely solely on changes in exchange rates. Instead, it is essential to enhance productive efficiency, strengthen the quality of investment, and encourage balanced capital accumulation. If domestic structural conditions remain weak, exchange rate adjustments alone cannot ensure long-term stability. Conversely, when productive capacity and stabilization mechanisms are robust, external adjustment supports sustainable economic growth.

Although the framework employed in this research has been deliberately kept stylized, this constitutes its primary strength, as it facilitates a clear understanding of the core mechanisms of adjustment. Future research could extend this model by incorporating policy shocks, expectations, stochastic disturbances, or empirical calibration tailored to developing economies. Such extensions would help bring theoretical findings closer to actual economic realities.

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